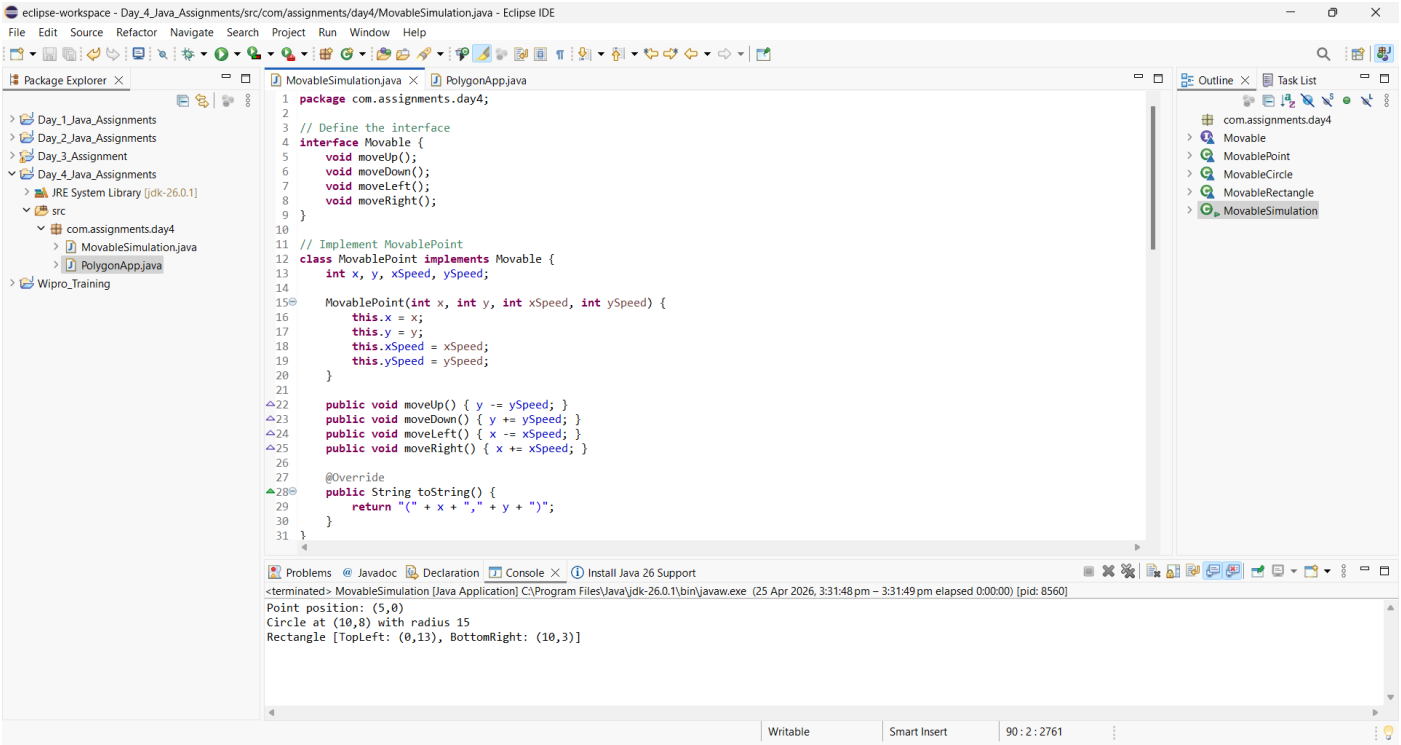
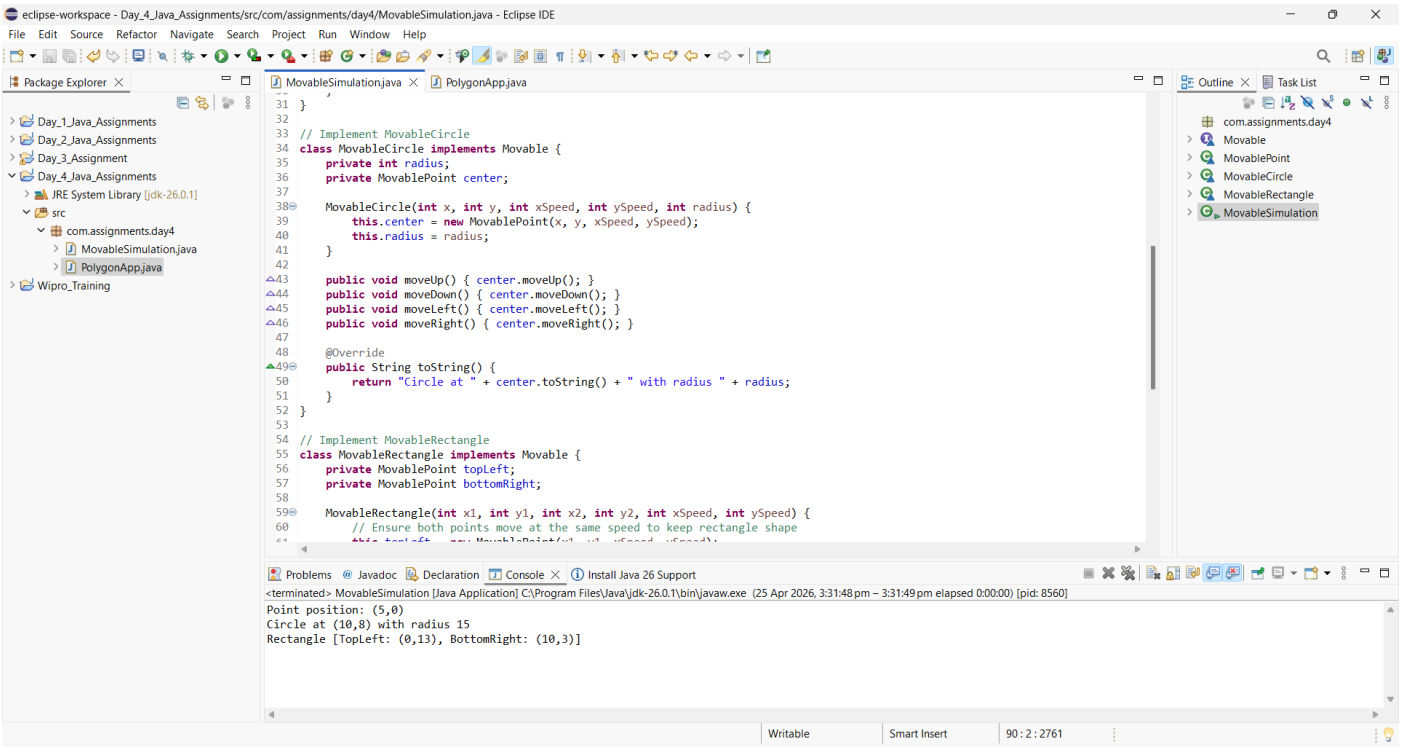
1. Moveable Shapes Simulation

- Define an interface Movable with methods: moveUp(), moveDown(), moveLeft(), moveRight().
- Implement classes: o MovablePoint(x, y, xSpeed, ySpeed) implements Movable o MovableCircle(radius, center: MovablePoint)

o MovableRectangle(topLeft: MovablePoint, bottomRight: MovablePoint) (ensuring both points have same speed)

- Provide toString() to display positions.
- In main(), create a few objects and call move methods to simulate motion.





eclipse-workspace - Day_4_Java_Assignments/src/com/assignments/day4/MovableSimulation.java - Eclipse IDE

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Package Explorer

- Day_1_Java_Assignments
- Day_2_Java_Assignments
- Day_3_Assignment
- Day_4_Java_Assignments
  - JRE System Library [jdk-26.0.1]
  - src
    - com.assignments.day4
      - MovableSimulation.java
      - PolygonApp.java
- Wipro_Training

MovableSimulation.java | PolygonApp.java

```java
        // Ensure both points move at the same speed to keep rectangle shape
        this.topLeft = new MovablePoint(x1, y1, xSpeed, ySpeed);
        this.bottomRight = new MovablePoint(x2, y2, xSpeed, ySpeed);
    }

    public void moveUp() { topLeft.moveUp(); bottomRight.moveUp(); }
    public void moveDown() { topLeft.moveDown(); bottomRight.moveDown(); }
    public void moveLeft() { topLeft.moveLeft(); bottomRight.moveLeft(); }
    public void moveRight() { topLeft.moveRight(); bottomRight.moveRight(); }

    @Override
    public String toString() {
        return "Rectangle [TopLeft: " + topLeft + ", BottomRight: " + bottomRight + "]";
    }
}

public class MovableSimulation {
    public static void main(String[] args) {
        Movable point = new MovablePoint(0, 0, 5, 5);
        point.moveRight();
        System.out.println("Point position: " + point);

        Movable circle = new MovableCircle(10, 10, 2, 2, 15);
        circle.moveUp();
        System.out.println(circle);

        Movable rect = new MovableRectangle(0, 10, 10, 0, 3, 3);
        rect.moveDown();
        System.out.println(rect);
    }
}
```

Outline | Task List

- com.assignments.day4
  - Movable
  - MovablePoint
  - MovableCircle
  - MovableRectangle
  - MovableSimulation

Problems | Javadoc | Declaration | Console | Install Java 26 Support

<terminated> MovableSimulation [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (25 Apr 2026, 3:31:48 pm – 3:31:49 pm elapsed 0:00:00) [pid: 8560]

```
Point position: (5,0)
Circle at (10,8) with radius 15
Rectangle [TopLeft: (0,13), BottomRight: (10,3)]
```

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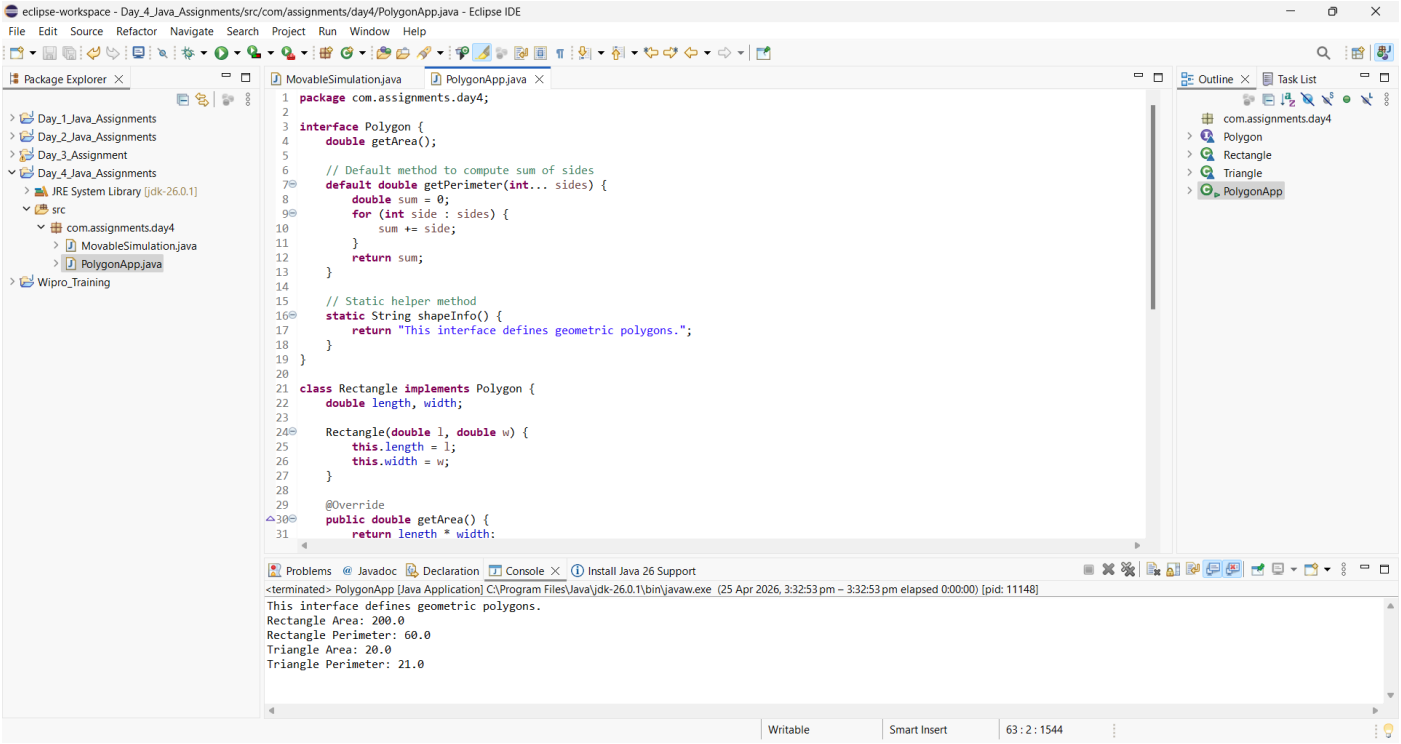
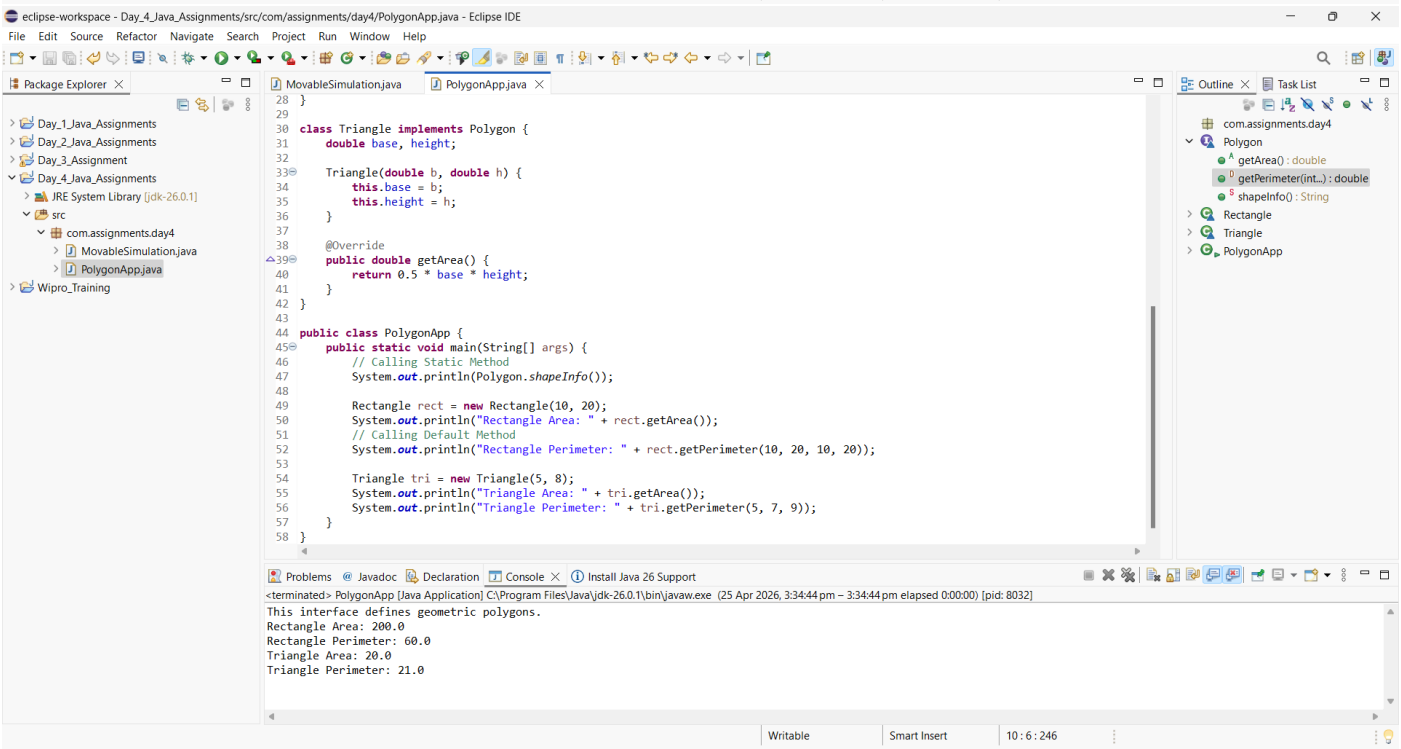
2. Default and Static Methods in Interfaces

- Declare interface Polygon with: o double getArea()

o default method default double getPerimeter(int... sides) that computes sum of sides

o a static helper static String shapeInfo() returning a description string

- Implement classes Rectangle and Triangle, providing appropriate getArea().
- In Main, call getPerimeter(...) and Polygon.shapeInfo().